

TASK 2: Cryptocurrency Portfolio Analysis

Performance, Volatility & Risk Metrics Evaluation (12-Month Period)

Student Name: Ayub Abdullahi Ali
Student ID: CA/DF1/247686

Domain: Finance & Investment Analysis
Organization: CodeAlpha

1. Executive Summary

This report presents a comprehensive risk and performance analysis of three primary digital assets—**Bitcoin (BTC)**, **Ethereum (ETH)**, and **Solana (SOL)**—over a 12-month trailing window. By examining key parameters including price trajectory, annualized volatility, network dominance, and systemic risk profiles, this paper establishes a research-backed asset allocation matrix tailored for balanced institutional and retail investment portfolios.

2. 12-Month Comparative Risk & Performance Matrix

Metric / Asset	Bitcoin (BTC)	Ethereum (ETH)	Solana (SOL)
Asset Classification	Digital Reserve / Macro Store	Smart Contract Platform (L1)	High-Speed Layer 1 Network
Current Price Range	~\$77,000 – \$78,000	~\$2,500 – \$2,520	~\$100 – \$102
12-Month High / Low	\$117,570 / \$60,210	\$4,830 / \$1,576	\$238.50 / \$66.75
Annualized Volatility	Moderate (~45%)	High (~60%)	Very High (~85%)
Market Dominance	~56.0%	~14.0%	~3.5%
Primary Catalyst	Spot ETF Flows & Halving	DeFi & Staking Adoption	High-Frequency Trading / DePIN

3. Individual Asset Deep-Dive Analysis

A. Bitcoin (BTC) – Macro Hedge & Reserve Asset

- **Trend Analysis:** Maintained a dominant market position, driven by sustained spot ETF inflows and expanding corporate treasury allocations before stabilizing around the \$77,000 level.
- **Risk Profile:** Lowest downside volatility within the digital asset class. Serves as a primary "risk-off" instrument in crypto portfolio construction.
- **Strategic Outlook:** Institutional adoption and fixed supply mechanics reinforce its position as digital gold.

B. Ethereum (ETH) – Smart Contract & Tokenization Settlement Layer

- **Trend Analysis:** Displayed solid performance in early market cycles before consolidating. Continues to lead in total value locked (TVL) and institutional staking.
- **Risk Profile:** Moderate-to-high volatility. Layer-2 migration creates execution efficiency while fragmenting immediate base-layer fee accrual.
- **Strategic Outlook:** Essential infrastructure core for Decentralized Finance (DeFi) and Real-World Asset (RWA) tokenization.

C. Solana (SOL) – High-Throughput Tactical Asset

- **Trend Analysis:** Exhibited high-beta performance, surging to local peaks before entering consolidation. Demonstrates strong consumer DApp transaction volume.
- **Risk Profile:** High volatility and larger peak-to-trough drawdowns. Optimized for throughput, requiring active risk monitoring.
- **Strategic Outlook:** Provides high growth/alpha potential in expanding market cycles.

4. Portfolio Diversification & Risk Allocation Strategies

Analyzing historical correlation indicates that Bitcoin and Ethereum remain strongly correlated (~ 0.82), while Solana provides high-beta upside capture. Combining these assets under structured allocation frameworks optimizes the Sharpe ratio while mitigating downside drawdown.

Recommended Model Portfolios:

- **Conservative Allocation:** 70% BTC | 25% ETH | 5% SOL (*Focus: Capital Preservation*)
- **Balanced Growth Allocation:** 50% BTC | 35% ETH | 15% SOL (*Focus: Optimal Risk-Adjusted Growth*)
- **Aggressive Alpha Allocation:** 35% BTC | 35% ETH | 30% SOL (*Focus: High Beta & Capital Appreciation*)

